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Investigation of the effect of cyclic borate ester groups in acrylic copolymers on paint and varnish coatings

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Abstract

In this study, cyclic borate ester-bearing methacrylic monomers were created and employed to synthesize acrylic copolymers with varying boron acrylate monomer contents (5, 10, and 20%). The monomers, methyl methacrylate, butyl acrylate, acrylic acid, and boron acrylate, were used to synthesize all of the polymers using free radical polymerization in a solvent medium. The polymers were characterized using FTIR (Fourier transform infrared spectroscopy), DSC (differential scanning calorimetry), Gel permeation chromatography (GPC), 1H and 11B NMR (proton and boron nuclear magnetic resonance), and HPLC. Metal surfaces, such as sheet metal, galvanized steel, and aluminum, were covered with varnish and paint compositions that had a fixed thickness. Contact angle value, glossiness, hardness, drying time, pot life, yellowing resistance, and gloss loss following UV were all measured to assess the coatings' physical characteristics. According to the findings, adding more borate ester groups to polymers increased their hardness as well as decreased the drying time of the coatings. When compared to the binders used in commercially available paints, the polymer with 10% boron acrylate monomer can be utilized to produce paints and varnishes with equal or superior physical properties.

Keywords

• Acrylic coatings , Borate ester , Boron acrylate copolymers , Paint and varnish , Road marking paints

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